

BM and TRANSITION SEMIGROUPS

We can associate with a BM (and any Markov process) two families of linear operators. The (transition) semigroup

$$P_t u(x) = E^x u(B_t) \quad u \in \mathcal{B}_b(\mathbb{R}^d), \quad x \in \mathbb{R}^d, \quad t \geq 0$$

and the resolvent

$$U_\alpha u(x) = E^x \int_0^\infty e^{-\alpha t} u(B_t) dt \quad u \in \mathcal{B}_b(\mathbb{R}^d), \quad x \in \mathbb{R}^d, \quad \alpha > 0.$$

In this section we will always assume that $(B_t)_{t \geq 0}$ is a BM adapted to $(\mathcal{F}_t)_{t \geq 0}$. We will assume that $\mathcal{F}_t$ is right continuous, i.e., $\mathcal{F}_t = \mathcal{F}_{t+}$ (example: $\overline{\mathcal{F}}_t^B$).

The semigroup.

A semigroup $(P_t)_{t \geq 0}$ on a Banach space $(\mathcal{B}, \|\cdot\|)$ is a family of linear operators

$$P_t : \mathcal{B} \rightarrow \mathcal{B}$$

s.t.

- $P_t P_r = P_{t+r}$ (Associativity)
- $P_0 = \text{id}$ (Identity element)

If we ask for the existence of an inverse element, then the family $(P_t)_{t \geq 0}$ would be a group. In our case it is just a semigroup.

We will consider semigroups on the Banach spaces

$\mathcal{B}_b(\mathbb{R}^d) :=$ family of bounded Borel measurable functions

$$f: \mathbb{R}^d \rightarrow \mathbb{R}, \text{ with } \|f\| = \|f\|_\infty$$

$C_0(\mathbb{R}^d) :=$ family of continuous functions vanishing at infinity ($\lim_{|x| \rightarrow \infty} u(x) = 0$), with $\|u\| = \|u\|_\infty$.

Lemma. The operator $P_t u(x) := \int \delta^x u(B_t)$ defines a semigroup of operators on $\mathcal{B}_b(\mathbb{R}^d)$.

Line. & linearity:

$$\begin{aligned} P_t(u_1 + u_2)(x) &= \int \delta^x (u_1(B_t) + u_2(B_t)) \\ &= P_t u_1(x) + P_t u_2(x) \end{aligned}$$

$$\bullet P_t: \mathcal{B}_b \rightarrow \mathcal{B}_b$$

$$\|P_t u(x)\| = \left\| \int \frac{1}{\sqrt{2\pi t}^d} e^{-\frac{(x-y)^2}{2t}} u(y) dy \right\|$$

$$\leq \|u\|_\infty \quad \text{Boundedness!}$$

$$x \mapsto \int \frac{1}{\sqrt{2\pi t}^d} e^{-\frac{(x-y)^2}{2t}} u(y) \quad \text{measurable!}$$

- $P_t P_s = P_{t+s}$

$$\begin{aligned}
 P_{t+s} \mu(n) &= \tilde{E}^n \mu(B_{t+s}) = \tilde{E}^n \tilde{E}^n [\mu(B_{t+s}) | \mathcal{F}_s] \\
 &= \tilde{E}^n \tilde{E}^{B_s} \mu(B_t) \\
 &= \tilde{E}^n P_t \mu(B_s) \\
 &= P_s P_t \mu(n)
 \end{aligned}$$

- $P_0 = 1 \quad \tilde{E}^n \mu(B_0) = \mu(n)$



Among various semigroups it will be useful to consider the so-called strongly continuous semigroup, i.e., such that

$$\|P_t \mu - \mu\| \rightarrow 0, \text{ for any } \mu \in \mathcal{P}$$

this implies continuity of the mapping

$$[0, \infty) \ni t \mapsto P_t \mu \in \mathcal{P}.$$

Indeed, suppose $h \downarrow 0$,

$$\begin{aligned}
 \|T_{t+h} \mu - T_t \mu\| &= \|T_t (T_h \mu - \mu)\| \\
 &\leq \|T_h \mu - \mu\| \xrightarrow{h \rightarrow 0} 0
 \end{aligned}$$

$$\|T_{t+h} \mu - T_t \mu\| = \|T_{t+h} (\mu - T_h \mu)\| \leq \|\mu - T_h \mu\| \xrightarrow{h \rightarrow 0} 0.$$

This property is related (in the case $\mathcal{F} = C_\infty(\mathbb{R}^d)$) with uniform stochastic continuity.

Lemma. P_t^x is uniformly stochastically continuous, i.e.,

$$\lim_{t \rightarrow 0} \sup_{x \in \mathbb{R}^d} P^x(|B_t - x| > \delta) = 0$$
for every $\delta > 0$.

Proof. $P^x(|B_t - x| > \delta) = P(|B_t|^2 > \delta^2)$
 Chebys $\leq \frac{\sigma |B_t|^2}{\delta^2} = \frac{td}{\delta^2}$

Here are the properties of the transition semigroup of BM.
 Proposition. For all $t \geq 0$.

- P_t is conservative: $P_t 1 = 1$
- P_t is a contraction on $\mathcal{B}_b$: $\|P_t u\|_\infty \leq \|u\|_\infty$
- P_t is positivity preserving: $u \in \mathcal{B}_b, u \geq 0 \rightarrow P_t u \geq 0$
- P_t is sub-Markovian: $u \in \mathcal{B}_b, 0 \leq u \leq 1 \rightarrow 0 \leq P_t u \leq 1$
- P_t has the Feller property: if $u \in C_\infty$ then $P_t u \in C_\infty$
- P_t is strongly continuous on C_∞ : $\lim_{t \rightarrow 0} \|P_t u - u\| = 0 \quad \forall u \in C_\infty$
- P_t has the strong Feller property: if $u \in \mathcal{B}_b$ then $P_t u \in C_b$.

continuous
bounded

Note: a) - d) Markov semigroup
 b) - d) sub-Markov
 b) - f) Feller
 b) - d), g) strong Feller

Proof.

$$a) P_t 1(x) = \mathbb{E}^x 1_{\mathbb{R}^d}(B_t) = P^x(B_t \in \mathbb{R}^d) = 1 \quad \forall x \in \mathbb{R}^d$$

$$b) \|P_t u(x)\|_{\infty} = \sup_x |P_t u(x)| \\ \leq \sup_x \mathbb{E}^x |u(B_t)| \\ \leq \|u\|_{\infty}$$

$$c) \text{ If } u \geq 0, \text{ then } P_t u(x) = \underbrace{\int u(y) P^x(B_t \in dy)}_{\geq 0}$$

$$d) \int_{\mathbb{R}^d} P^x(B_t \in dy) \leq \int_{\mathbb{R}^d} \underbrace{u(y)}_{0 \leq u \leq 1} P^x(B_t \in dy) \leq \int_{\mathbb{R}^d} \underbrace{1}_{=1} P^x(B_t \in dy)$$

e) $\int \phi_t d\mathbb{P}^n$

$$\mathbb{E}^n[u(B_t)] = \mathbb{E}^n[u(x+B_t)]$$

$$|u(x+B_t)| \leq \|u\|_\infty \quad \text{and} \quad \mathbb{E}^n[\|u\|_\infty] \rightarrow 0$$

hence by D.M. Conv.

$$\begin{aligned} \lim_{n \rightarrow \infty} \mathbb{E}^n[u(B_t)] &= \mathbb{E}^y[u(y+B_t)] \\ &= \mathbb{E}^y(u)(B_t) \end{aligned}$$

$$\text{and also} \quad \lim_{n \rightarrow \infty} \mathbb{E}^n[u(B_t)] = \lim_{n \rightarrow \infty} \mathbb{E}^n[u(x+B_t)] = 0$$

f) Let $\varepsilon > 0$. Since $u \in C_0(\mathbb{R}^d)$ is also unif. continuous then there is $\delta > 0$ s.t.

$$|u(x) - u(y)| < \varepsilon \quad \text{for all } |x - y| < \delta.$$

Thus

$$\|P_t u - u\|_\infty = \sup_{x \in \mathbb{R}^d} |\mathbb{E}^x[u(B_t)] - u(x)|$$

$$\leq \sup_x \mathbb{E}^x[|u(B_t) - u(x)|]$$

$$\leq \sup_x \int_{|B_t - x| < \delta} |u(B_t) - u(x)| dP^x +$$

$$\sup_x \int_{|B_t - x| \geq \delta} |u(B_t) - u(x)| dP^x$$

$$\leq \varepsilon \sup_x P^x(|B_t - x| < \delta) + 2\|u\|_\infty \sup_x P^x(|B_t - x| \geq \delta)$$

$$\leq \varepsilon + 24n\omega_\infty P^n(|B_t - u| \geq \delta)$$

hence $\limsup_{t \rightarrow \infty} \|P_t u - u\| \leq \varepsilon$ for any $\varepsilon > 0$ fixed

since $\sup_n P^n(|B_t - u| \geq \delta) \rightarrow 0$ as $t \rightarrow \infty$.

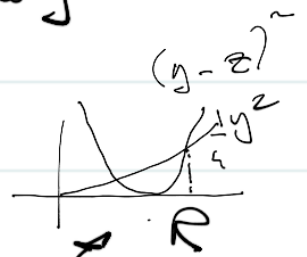
The claim follows as $\varepsilon \rightarrow 0$

g) let $u \in \mathcal{B}_b(\mathbb{R}^d)$. Then

$$P_t u(z) = \frac{1}{(2\pi t)^{d/2}} \int_{\mathbb{R}^d} u(y) e^{-\frac{|z-y|^2}{2t}} dy$$

Fix $R > 0$. For $|y| \geq R$ we have

$$|z-y|^2 \geq (|y| - |z|)^2 \geq \frac{1}{4}|y|^2$$



hence

$$u(y) e^{-\frac{|z-y|^2}{2t}} = \mathbf{1}_{B(0,R)}(y) u(y) e^{-\frac{|z-y|^2}{2t}} + \mathbf{1}_{B^c(0,R)}(y) u(y) e^{-\frac{|z-y|^2}{2t}}$$

For R big enough

$$\leq \|u\|_\infty \left(\mathbf{1}_{B(0,R)} + \mathbf{1}_{B^c(0,R)} e^{-\frac{|y|^2}{8t}} \right)$$

integrable for $y \in \mathbb{R}^d$

hence you can send $z \rightarrow u$ using dominated convergence.

Using semigroup notation it is possible to express the finite dimensional distribution of a Markov process. As we did for the 1-dim case, we can obtain "more" information (the finite dim) from "few" information (the semigroup representation) by using Markov property.

Theorem. Let $t_0 = 0, t_1, \dots, t_n$, $B_1, \dots, B_n \in \mathcal{B}(\mathbb{R}^d)$ and $x \in \mathbb{R}^d$. If $(P_t)_{t \geq 0}$ is the transition semigroup of a BM, then

$$P^n(B_{t_1} \in C_1, \dots, B_{t_n} \in C_n) = P_x \left[1_{C_1} P_{t_2-t_1} \left[1_{C_2} \dots P_{t_n-t_{n-1}} 1_{C_n} \right] \right]$$

Proof. Take $n=2$

$$\begin{aligned} P^n(B_{t_1} \in B_1, B_{t_2} \in B_2) &= E^n [1_{B_{t_1} \in B_1} 1_{B_{t_2} \in B_2}] \\ &= E^n \left[E^n [1_{B_{t_1} \in B_1} 1_{B_{t_2} \in B_2} | \mathcal{F}_{t_1}] \right] \\ &= E^n \left[1_{B_{t_1} \in B_1} E^n [1_{B_{t_2} \in B_2} | \mathcal{F}_{t_1}] \right] \\ &= E^n \left[1_{B_{t_1} \in B_1} E^{B_{t_1}} [1_{B_{t_2-t_1} \in B_2}] \right] \end{aligned}$$

$$= P_{t_1} \left[1_{B_1}^{(0)} P_{t_2-t_1} 1_{B_2}^{(1)} \right] (2)$$

Iterating these computations we obtain the result for $n > 2$.

Previously we saw how to construct a process, starting from a 1-dim marginal, using MP to obtain a finite dimensional and then constructing the process using Kolmogorov's procedure. Here we address the problem of finding a MP given a semigroup of operators.

If T_t is a Markov semigroup and $P_t(x; c) := T_t 1_c^{(x)}$ is a p.m.

$c \mapsto P_t(x, c)$ is a prob. measure
 $t \in \mathbb{R}_+ \mapsto P_t(x, c)$ is measurable

NB:
Assumptions!!

for all $x \in \mathbb{R}^d$, $c \in \mathcal{B}(\mathbb{R}^d)$, this procedure is possible. Indeed using the formula above we define a finite dimensional or:

$$P_{t_1, \dots, t_n}^{x_1, \dots, x_n}(c_1 \times \dots \times c_n) := T_{t_1} [1_{c_1} T_{t_2-t_1} [1_{c_2} \dots T_{t_n-t_{n-1}} 1_{c_n} \dots] x_1]$$

It can be seen that this prob. measure on $\mathcal{Q}^{d \times n}$ satisfies the consistency condition of Kolmogorov or theorem, e.g., check the properties:

$$P_{t_1 \dots t_n}^n (C_1 \times \dots \times \mathbb{R}^d \times \dots \times C_n)$$

$$= \tau_{t_1} [1_{C_1} \tau_{t_2 - t_1} [1_{C_2} \dots 1_{C_{n-1}} \underbrace{\tau_{t_n - t_{n-1}} 1_{\mathbb{R}^d} \tau_{t_{n+1} - t_n} 1_{C_{n+1}} \dots \tau_{t_m - t_{m-1}} 1_{C_m}}_{= \tau_{t_{n+1} - t_{n-1}} 1_{C_{n+1}}}]]_{(n)}$$

$$= P_{t_1 \dots t_{n-1}, t_{n+1}, \dots, t_m}^n (C_1 \times \dots \times C_{n-1} \times C_{n+1} \times \dots \times C_m).$$

We can therefore use Kolmogorov's construction to construct the process.

Now: Is the process $X = (X_t)_{t \geq 0}$ a Markov process?

Yes! For this, it is enough to show that

$$P_{t-s}(X_s, C) = P^n(X_t \in C | \mathcal{F}_s) \quad \begin{matrix} n \in \mathbb{N} \\ C \in \mathcal{B} \\ s < t \end{matrix}$$

l.e.

$$E^{x_0} 1_{[X_{t-r} \in C]} = P^n(X_t \in C | \mathcal{F}_r).$$

Hence we should prove that

$$* \quad \int_G 1_{[X_t \in C]} dP^x = \int_G P_{t-r}(x_r, C) dP^x$$

for every $G \in \mathcal{F}_r$. Since $\mathcal{F}_0$ is generated by the α -system $\mathcal{G}$

$$\{X_{\alpha_0} \in C_0, X_{\alpha_1} \in C_1, \dots, X_{\alpha_m} \in C_m\}, \quad \alpha_0 = 0 < \alpha_1 < \dots < \alpha_m = r$$

$$C_1, \dots, C_m \in \mathcal{B}(\mathbb{R}^d)$$

$$m \in \mathbb{N}$$

it is enough to show $*$ for G in $\mathcal{G}$.
We have that $*$ becomes

$$\int_{\Omega} \underbrace{1_{[X_{\alpha_1} \in C_1, \dots, X_{\alpha_m} \in C_m]}}_{\text{an element } G \in \mathcal{G}} P_{t-\alpha_m}(X_{\alpha_m}, C) dP^x \quad (*)$$

$\downarrow$
 $\text{since } \alpha = \alpha_m$

$$= \int_{\Omega} \underbrace{1_{[X_{\alpha_1} \in C_1, \dots, X_{\alpha_m} \in C_m]}}_{G \in \mathcal{G}} 1_{(X_t \in C)} dP^x$$

$$\textcircled{A} = \int \dots \int_{r_1 \dots r_m} p^n (dx_1 \dots dx_m) \mathbb{1}_{[x_1 \in C_1]} \dots \mathbb{1}_{[x_m \in C_m]} p_{t-r_m}(x_m, C) = *$$

i.e., taking the l.h.s.,

$$E^n \mathbb{1}_{[x_{r_1} \in C_1]} \mathbb{1}_{[x_{r_2} \in C_2]} \dots \mathbb{1}_{[x_{r_m} \in C_m]} p_{t-r_m}(x_{r_m}, C) =$$

$$* = \int p^n_{r_1 \dots r_{m-1}} (C_1 x \dots x_{C_{m-1}} x_{C_m}) \mathbb{1}_{[x_m \in C_m]} p_{t-r_m}(x_m, C)$$

by def of $p^n_{r_1 \dots r_m}$

$$= T_{r_1} \left[\mathbb{1}_{C_1} T_{r_2-r_1} \left[\mathbb{1}_{C_2} \dots \mathbb{1}_{C_{m-1}} T_{r_m-r_{m-1}} \left[\mathbb{1}_{C_m} p_{t-r_m}(\cdot, C) \right] \right] \right] (x)$$

$\underbrace{\quad}_{T_{t-r_m} \mathbb{1}_C}$

~~~~~

$$= T_{r_1} \left[ \mathbb{1}_{C_1} T_{r_2-r_1} \left[ \mathbb{1}_{C_2} \dots \mathbb{1}_{C_{m-1}} T_{r_m-r_{m-1}} \left[ \mathbb{1}_{C_m} T_{t-r_m} \mathbb{1}_C \right] \right] \right] (x)$$

By def of  $p^n_{r_1 \dots r_m}(C_1 \dots C_m)$

$$= p^n_{r_1 \dots r_m, t}(C_1 x \dots x_{C_{m-1}} x_{C_m} x_C)$$

$$= E^n \mathbb{1}_{[x_{r_1} \in C_1 \dots, x_{r_m} \in C_m, x_t \in C]}$$

= r.h.s of  $\textcircled{A}$ .

What we have now for Feller semigroups?

- Riesz Representation theorem.

Let  $\Lambda$  be a positive linear functional on  $C_c(\mathbb{R}^d)$ .  
Then there exists a  $\sigma$ -algebra  $\mathcal{M}$  on  $\mathbb{R}^d$  s.t.  $\mathcal{M}$  contains all the Borel sets of  $\mathbb{R}^d$  and there exists a unique positive measure  $\mu$  on  $\mathcal{M}$  s.t.

$$\Lambda f = \int_{\mathbb{R}^d} f d\mu$$

for every  $f \in C_c(\mathbb{R}^d)$  and has the following additional properties:

- $\mu(K) < +\infty$  for every compact  $K \subset \mathbb{R}^d$
- For every  $\sigma \in \mathcal{M}$  we have
$$\mu(\sigma) = \sup \{ \mu(V) : \sigma \subset V, V \text{ open} \}$$
- The relation  $\mu(\sigma) = \sup \{ \mu(K) : K \subset \sigma, K \text{ compact} \}$  holds for every open set  $\sigma$ , and for every  $\sigma \in \mathcal{M} : \mu(\sigma) < +\infty$
- If  $\sigma \in \mathcal{M}$ ,  $A \subset \sigma$  and  $\mu(\sigma) < \infty$  then  $A \in \mathcal{M}$ .

Recall that  $C_c(\mathbb{R}^d)$  is dense in  $C_0(\mathbb{R}^d)$ . Hence, take  $f \in C_0(\mathbb{R}^d)$ , we have that there exists  $f_n \in C_c(\mathbb{R}^d)$  s.t.

$\|f_n - f\| \rightarrow 0$ . Furthermore,

$$\Lambda f_n = \int f_n d\mu \quad f_n \in C_c(\mathbb{R}^d)$$

We recall that

~~Then~~  $\mu(\mathbb{R}^d) = \sup \left\{ \int h d\mu : 0 \leq h \leq 1, h \in C_c(\mathbb{R}^d) \right\}.$

↳ this can be proved

Take  $\lambda$  bounded in  $C_0(\mathbb{R}^d)$ , then  $\| \lambda \| \leq C \| \lambda \|$  (suppose  $C=1$ ). ↗ assumed for Riesz representation!

We have that  $\lambda$  is integral on  $C_c(\mathbb{R}^d)$  and using ④ we have  $\mu(\mathbb{R}^d) \leq 1$  (since  $\| \lambda \| \leq 1$  on  $C_c(\mathbb{R}^d)$ ).

Therefore

$$\lambda(f) = \int f d\mu \quad f \in C_0(\mathbb{R}^d)$$

defines a bounded linear functional on  $C_0(\mathbb{R}^d)$ .

Now, take  $f_n \rightarrow f$  in  $C_0(\mathbb{R}^d)$ , we have

$$\left\| \int f_n d\mu - \int f d\mu \right\| \leq \int \| f_n - f \| d\mu$$

bounded in  $n$  since  $f_n$  is bounded and then bounded

$$\| f_n - f \| \leq \text{const}(f)$$

$$\rightarrow 0 \quad \text{as } n \rightarrow \infty$$

↳ by DCT.

Since  $\lambda : C_0 \rightarrow \mathbb{R}$  is bounded, it is also continuous, in  $C_0(\mathbb{R}^d)$  and thus

$$\lambda f_n \rightarrow \lambda f \quad \text{in } C_0(\mathbb{R}^d).$$

By uniqueness of limits we have  $\lambda f = \int f d\mu$  and also that  $\mu(\mathbb{R}^d) \leq 1$ .

If  $T_t$  is a Feller semigroup (on  $C_b(\mathbb{R}^d)$ ), i.e., (b-f)  
 we don't need to assume that

$$T_t u = \int u(y) P_t(x, dy)$$

Indeed this is a consequence of Riesz representation theorem:

There exists a unique kernel s.t.

$$T_t u(x) = \int u(y) P_t(x, dy) \quad u \in C_b(\mathbb{R}^d)$$

and  $P_t(x, \mathbb{R}^d) \leq 1$  and  
 $C \mapsto P_t(x, C)$  is prob. meas &  $\mu$

Hence we should repeat the previous argument, taking  
 into account that  $P_t(x, C) \leq 1$  and that  
 $\hookrightarrow$  not =

we must prove  $\forall (t, x) \mapsto P_t(x, C) \in \mathcal{B}(\mathbb{R}^d)$   
 is measurable. This last assertion is a consequence of  $\textcircled{A}$   
~~the regularity of the Riesz measure~~ To deal with  
 the case

$$P_t(x, \mathbb{R}^d) < 1 \quad \text{here is a trick.}$$

$\textcircled{A}$  strong continuity

Consider the one-point compactification of  $\mathbb{R}^d$ , i.e., we add the point  $\partial \in \mathbb{R}^d$ , and we define

$$p_t^{\partial}(x, \{\partial\}) := 1 - p_t(x, \mathbb{R}^d) \quad x \in \mathbb{R}^d, t \geq 0$$

$$p_t^{\partial}(\partial, c) := \delta_{\partial}(c)$$

$$c \in \mathcal{C}(\mathbb{R}^d \cup \{\partial\}) \quad t \geq 0$$

$$p_t^{\partial}(x, c) := p_t(x, c)$$

$$x \in \mathbb{R}^d, c \in \mathcal{C}(\mathbb{R}^d), t \geq 0$$

→  $\partial$  is absorbing

At this point one can construct the process on  $\mathbb{R}^d \cup \{\partial\}$  ...

## The generator

Let  $\phi: [0, \infty) \rightarrow \mathbb{R}$  be a continuous function which satisfies the functional equation

$$\phi(t) \phi(r) = \phi(t+r) \quad \phi(0) = 1.$$

then

$$\phi(t) = e^{at}$$

and

$$a = \frac{d}{dt} \phi(t) \Big|_{t=0} = \lim_{t \rightarrow 0} \frac{\phi(t) - 1}{t}.$$

The idea is that, since  $P_t P_r = P_{t+r}$ , we can write, in an appropriate sense, that

$$P_t = e^{At}$$

for some operator  $A$  on  $\mathcal{B}(C_\infty(\mathbb{R}^d))$  (in our case).

**Definition.** Let  $(P_t)_{t \geq 0}$  be a Feller semigroup on  $C_\infty(\mathbb{R}^d)$ . Then let

$$Au := C_\infty(\mathbb{R}^d) - \lim_{t \rightarrow 0} \frac{P_t u - u}{t}$$

and

$$\mathcal{D}(A) := \left\{ u \in C_\infty(\mathbb{R}^d) : \exists g \in C_\infty(\mathbb{R}^d) : \lim_{t \rightarrow 0} \left\| \frac{P_t u - u}{t} - g \right\|_\infty = 0 \right\}$$



Then  $(A, \mathcal{D}(A))$  is called the generator of the semigroup  $(P_t)_{t \geq 0}$ .

The operator  $A$  so defined is linear on  $A: \mathcal{D}(A) \rightarrow C_\infty(\mathbb{R}^d)$ .

Let us compute the generator of  $BM^d$  and its domain.

Example. For a  $BM^d$ , we have that

$$C_\infty^2(\mathbb{R}^d) \subset \mathcal{D}(A) \text{ and } A|_{C_\infty^2(\mathbb{R}^d)} = \frac{1}{2} \Delta$$

where

$$C_\infty^2(\mathbb{R}^d) := \{u \in C_\infty(\mathbb{R}^d) : \partial_j u, \partial_j \partial_k u \in C_\infty(\mathbb{R}^d) \text{ for } j, k = 1, \dots, d\}.$$

→ Taylor for  $f(u+y)$  in  $y=0$ . Hence

$$f(u+y) = f(u) + \sum_{j=1}^d \partial_{x_j} f(u) y_j + \sum_{j,k=1}^d \frac{1}{2} \partial_{x_j} \partial_{x_k} f(u) y_j y_k$$

Take  $u \in C_\infty^2(\mathbb{R}^d)$ . Take

$$\left| E \left[ \frac{u(B_t + u) - u(u)}{t} - \frac{1}{2} \sum_{j=1}^d \partial_{x_j}^2 u(u) \right] \right|$$

take  $\xi_t = \xi(u, t, \omega)$  between  $u$  and  $u+B_t(\omega)$

and use Taylor

$$= \left| E \left[ \frac{1}{t} \left[ \cancel{u(u)} + \sum_{j=1}^d \partial_{x_j} u(u) B_t^j + \frac{1}{2} \sum_{j,k=1}^d \partial_{x_j} \partial_{x_k} u(\xi_t) B_t^j B_t^k - \cancel{u(u)} \right] - \frac{1}{2} \sum_{j=1}^d \partial_{x_j}^2 u(u) \right] \right|$$

$$= \left| E \left[ \frac{1}{t} \sum_{j=1}^d \partial_j u(n) B_t^j + \frac{1}{2t} \sum_{j,k=1}^d \partial_j \partial_k u(\xi_t) B_t^j B_t^k - \frac{1}{2} \sum_{j=1}^d \partial_j^2 u(n) \right] \right|$$

$$E B_t^j = 0 \quad E B_t^j B_t^k = \begin{cases} t & j=k \\ 0 & j \neq k \end{cases}$$

is cancelled for  $E B_t^j = 0$

$$= \frac{1}{2} \left| \frac{1}{t} E \left[ \sum_{j,k=1}^d \partial_j \partial_k u(\xi_t) B_t^j B_t^k - \sum_{j=1}^d \partial_j^2 u(n) B_t^j B_t^j \right] \right|$$

$$= \frac{1}{2t} E \left[ \sum_{j,k=1}^d (\partial_j \partial_k u(\xi_t) - \partial_j \partial_k u(n)) B_t^j B_t^k \right]$$

Recall that, by Taylor formula, we have that

per la media delle derivazioni

$$\frac{1}{2} \partial_j \partial_k u(\xi_t) = \int_0^1 \partial_j \partial_k u(n + r B_t) (1-r) dr$$

so we can see that  $\partial_j \partial_k u(\xi_t(\omega))$  is measurable.  
Use now Cauchy Schwartz

$$(E XY)^2 \leq E X^2 E Y^2 \rightarrow E XY \leq \sqrt{E X^2} \sqrt{E Y^2}$$

$$\left( \sum_{i=1}^n u_i v_i \right)^2 \leq \sum_{i=1}^n u_i^2 \sum_{i=1}^n v_i^2 \rightarrow \sum_{i=1}^n u_i v_i \leq \sqrt{\sum_{i=1}^n u_i^2} \sqrt{\sum_{i=1}^n v_i^2}$$

$$\leq \frac{1}{2} \left| E \left[ \left( \sum_{j,k=1}^d (\partial_j \partial_k u(\xi_t) - \partial_j \partial_k u(n)) B_t^j B_t^k \right)^2 \right]^{1/2} \cdot \left( \sum_{j,k=1}^d \frac{(B_t^j B_t^k)^2}{t^2} \right)^{1/2} \right|$$

it depends from the parts that is

DL KROVO C-5

$$\leq \frac{1}{2} \left( \sum_{j,k=1}^d \left( \partial_j \partial_k u(\xi_t) - \partial_j \partial_k u(n) \right)^2 \right)^{1/2} \left( \sum_{j,k=1}^d \left( \frac{B_t^j B_t^k}{t^2} \right)^2 \right)^{1/2}$$

$$\sum_{j,k=1}^d (B_t^j)^2 (B_t^k)^2 = \sum_{j=1}^d (B_t^j)^2 \sum_{k=1}^d (B_t^k)^2 = |B_t|^2 |B_t|^2$$

$$= \frac{1}{2} \left( \sum_{j,k=1}^d \left( \partial_j \partial_k u(\xi_t) - \partial_j \partial_k u(n) \right)^2 \right)^{1/2} \left( \frac{\sum |B_t|^4}{t^2} \right)^{1/2}$$

$$\sum |B_t|^4 = t^2 \sum |B_1|^4 \rightarrow \infty$$

$$= \frac{1}{2} \sqrt{\sum |B_1|^4} \left( \sum_{j,k=1}^d \left( \partial_j \partial_k u(\xi_t) - \partial_j \partial_k u(n) \right)^2 \right)^{1/2}$$

$$\leq (2\|u\|)^2 \text{ for any } j, k.$$

$$\frac{np}{n} \left| \sum \left[ \frac{u(B_t + \Delta) - u(n)}{t} - \frac{1}{2} \sum_{j=1}^d \partial_j^2 u(n) \right] \right|$$

$$\leq \frac{1}{2} \sqrt{\sum |B_1|^4} \frac{np}{n} \left( \sum_{j,k=1}^d \left( \partial_j \partial_k u(\xi_t) - \partial_j \partial_k u(n) \right)^2 \right)^{1/2}$$

$$\leq \frac{1}{2} \sqrt{\sum |B_1|^4} \left( \sum_{j,k=1}^d \left( \frac{np}{n} \left| \partial_j \partial_k u(\xi_t) - \partial_j \partial_k u(n) \right| \right)^2 \right)^{1/2}$$

$$\leq 2\|u\|$$

By DOM CONV

$$t \rightarrow 0 \rightarrow \frac{1}{2} \sqrt{\delta(B)}^n \left( \sum_{j,k=1}^d \underbrace{\delta \left( \lim_{t \rightarrow 0} \frac{1}{n} \left| \partial_j \partial_k u(\xi_t) - \partial_j \partial_k u(x) \right| \right)^2}_{\rightarrow 0 \text{ since}} \right)^{\frac{1}{2}}$$

$\partial_j \partial_k u(x)$  is  $C^\infty(\mathbb{R}^d)$   
and thus uniformly  
continuous and  
 $\xi_t \rightarrow x, t \rightarrow 0$ .

thus prove that on  $C_c^\infty(\mathbb{R}^d)$  the generator  
of the Brownian semigroup is  $\Delta$ . ]

It turns out that the mapping

$$[0, \infty) \ni t \mapsto P_t u \in C_0(\mathbb{R}^d)$$

can be viewed as a solution to a Cauchy problem  
on  $C_0(\mathbb{R}^d)$ . In order to deal with this object  
we should be able to integrate functions of the  
form

$$q: [0, \infty) \mapsto \mathbb{R}$$

↳ Banach space.

This requires the notion of Bochner integral. We will  
avoid the use of Bochner integral or Fubini.

Since each operator  $(P_t)_{t \geq 0}$  is given by a family of measurable kernels, suppose we need to define

$$L := \int_0^t P_r \, dr$$

i.e.,

$$L u = \int_0^t P_r u \, dr$$

Then we say that

$$L u(x) := \int_0^t \int u(y) P_r(x, dy) \, dr.$$

Here is the fundamental relation between generators and semigroups.

**Lemma.** Let  $(P_t)_{t \geq 0}$  be a Feller semigroup with generator  $(A, \mathcal{D}(A))$ . It is true that

$$a) \quad \frac{d}{dt} P_t u = A P_t u = P_t A u \quad u \in \mathcal{D}(A), t \geq 0$$

↳ strong derivative

$$b) \quad \text{For all } u \in C_0(\mathbb{R}^d) \text{ we have } \int_0^t P_r u \, dr \in \mathcal{D}(A)$$

and

Proof Solution

$$P_t u - u = A \int_0^t P_r u \, dr \quad u \in C_0(\mathbb{R}^d), t \geq 0$$

$$= \int_0^t A P_r u \, dr \quad u \in \mathcal{D}(A), t \geq 0$$

$$= \int_0^t P_s A u \, ds \quad u \in \mathcal{D}(A), t \geq 0.$$

aim

a) let  $0 < \varepsilon < t$  and  $u \in \mathcal{D}(A)$ . We prove that

$$\left\| \frac{P_{t+\varepsilon} u - P_t u}{\varepsilon} - P_t A u \right\|_\infty \rightarrow 0 \text{ as } \varepsilon \rightarrow 0$$

and that

$$\left\| \frac{P_t u - P_{t-\varepsilon} u}{\varepsilon} - P_t A u \right\|_\infty \rightarrow 0 \text{ as } \varepsilon \rightarrow 0$$

i.e.  $\frac{d}{dt} P_t u = P_t A u$  (Any derivative).

$$\left\| \frac{P_{t+\varepsilon} u - P_t u}{\varepsilon} - P_t A u \right\|_\infty = \left\| \frac{P_\varepsilon P_t u - P_t u}{\varepsilon} - P_t A u \right\|_\infty$$

$$= \left\| P_t \frac{P_\varepsilon u - u}{\varepsilon} - P_t A u \right\|_\infty \leq \left\| \frac{P_\varepsilon u - u}{\varepsilon} - A u \right\|_\infty \xrightarrow{\varepsilon \rightarrow 0} 0$$

$\hookrightarrow$  by def of  $A$ .

Similarly

$$\begin{aligned} \left\| \frac{P_t u - P_{t-\varepsilon} u}{\varepsilon} - P_t A u \right\|_\infty &= \left\| P_{t-\varepsilon} \frac{P_\varepsilon u - u}{\varepsilon} - P_{t-\varepsilon} A u \right. \\ &\quad \left. + P_{t-\varepsilon} A u - P_t A u \right\|_\infty \end{aligned}$$

$$\leq \left\| P_{t-\varepsilon} \frac{P_{t-\varepsilon} u - u}{\varepsilon} - P_{t-\varepsilon} A u \right\| + \left\| P_{t-\varepsilon} A u - P_{t-\varepsilon} P_{\varepsilon} A u \right\|$$

$$\leq \underbrace{\left\| \frac{P_{t-\varepsilon} u - u}{\varepsilon} - A u \right\|}_{\rightarrow 0 \text{ by def of } A} + \underbrace{\left\| A u - P_{\varepsilon} A u \right\|}_{\rightarrow 0 \text{ by strong continuity of } P_{\varepsilon} u}.$$

$\rightarrow 0$  as  $\varepsilon \rightarrow 0$

$$\text{Hence } \frac{d}{dt} P_t u = \frac{d}{dt} P_t u = P_t A u.$$

$$\text{Now note that } \left\| P_{\varepsilon} - \frac{P_t - P_{t-\varepsilon}}{\varepsilon} P_t u - P_t A u \right\|$$

$$= \left\| P_t \left( \frac{P_{\varepsilon} - I}{\varepsilon} u - A u \right) \right\|$$

$$\leq \left\| \frac{P_{\varepsilon} - I}{\varepsilon} u - A u \right\| \rightarrow 0 \text{ as } \varepsilon \rightarrow 0.$$

i.e., since  $A := s\text{-}\lim_{\varepsilon \rightarrow 0} \frac{P_{\varepsilon} - I}{\varepsilon}$  (by def of  $A$ ) then we proved that

$$A P_t u = P_t A u$$

(and also  $P_t: \mathcal{D}(A) \rightarrow \mathcal{D}(A)$ ).

Let us prove b) now:



Take  $u \in \mathcal{B}_b(\mathbb{R}^d)$ ,  $t, \varepsilon > 0$ . Then by def of  $P_r$

$$\begin{aligned} P_\varepsilon \int_0^t P_r u(x) dr &= \int_0^t \int_0^t \int u(y) P_r(z, dy) dr P_\varepsilon(x, dz) \\ &= \int_0^t \iint u(y) P_r(z, dy) P_\varepsilon(x, dz) dr = \int_0^t \underbrace{P_r P_\varepsilon}_{P_{r+\varepsilon}} u(x) dr \\ &\quad \downarrow \text{Fubini} \end{aligned}$$

Hence

$$\frac{P_\varepsilon - I}{\varepsilon} \int_0^t P_r u(x) dr = \int_0^t P_{r+\varepsilon} u(x) dr - \int_0^t P_r u(x) dr$$

$$= \frac{1}{\varepsilon} \left[ \int_\varepsilon^{t+\varepsilon} P_r u(x) dr - \int_0^t P_r u(x) dr \right]$$

change variable  
can be interchanged

$0 \dots \varepsilon \dots t \dots t+\varepsilon$

$$= \frac{1}{\varepsilon} \left[ \int_t^{t+\varepsilon} P_r u(x) dr - \int_0^\varepsilon P_r u(x) dr \right] \quad \otimes$$

$$\frac{1}{\varepsilon} \int_r^{r+\varepsilon} P_r u(x) dr \Big|_{r=t, r=0}$$

Hence we prove that

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \int_r^{r+\varepsilon} P_{\sigma} u(x) dx = P_r u(x)$$

which means that

$$\int_0^t P_{\sigma} u(x) dx \in \mathcal{Q}(A)$$

and also, by (8),

$$A \int_0^t P_{\sigma} u(x) dx = P_t u - u.$$

We have that

$$\left| \frac{1}{\varepsilon} \int_r^{r+\varepsilon} P_{\sigma} u(x) dx - P_r u(x) \right|$$

$$\leq \frac{1}{\varepsilon} \int_r^{r+\varepsilon} |P_{\sigma} u(x) - P_r u(x)| dx$$

$$\leq \frac{1}{\varepsilon} \int_r^{r+\varepsilon} \sup_{t \in [r, r+\varepsilon]} \|P_{\sigma} u - P_r u\|_{\infty} dx$$

$$\leq \sup_{t \in [r, r+\varepsilon]} \|P_{\sigma} u - P_r u\|_{\infty} = \sup_{t \in [r, r+\varepsilon]} \|P_r (P_{\sigma-r} u - u)\|_{\infty}$$

$$\leq \sup_{t \in [r, r+\varepsilon]} \|P_{\sigma-r} u - u\|_{\infty} = \sup_{t \in [0, \varepsilon]} \|P_{\sigma} u - u\|_{\infty}$$

since  $\sigma \mapsto P_{\sigma} u$  is continuous then it is unif. cont. on

$[0, \varepsilon]$ : hence

$$\sup_{t \in [0, \varepsilon]} \|P_{\sigma} u - P_0 u\|_{\infty} \rightarrow 0.$$

Then take  $u \in \mathcal{D}(A)$ :

$$\begin{aligned} \int_0^t P_s A u(x) ds &= \int_0^t A P_s u(x) ds = \int_0^t \frac{d}{ds} P_s u(x) ds \\ &\stackrel{\text{by def. 2)}}{=} P_t u(x) - u(x) \\ &= A \int_0^t P_s u(x) ds \\ &\hookrightarrow \text{already proved.} \end{aligned}$$

We only state the following Corollary

Corollary. Let  $(P_t)_{t \geq 0}$  be a Feller semigroup with generator  $(A, \mathcal{D}(A))$ .

a)  $\mathcal{D}(A)$  is dense in  $C_0(\mathbb{R}^d)$

b)  $(A, \mathcal{D}(A))$  is closed i.e.

$$\left\{ \begin{array}{l} \{u_n\}_{n \geq 1} \subset \mathcal{D}(A), \lim_n u_n = u \\ \{A u_n\}_{n \geq 1} \text{ is a Cauchy sequence} \end{array} \right\}$$

$$\left\{ \begin{array}{l} u \in \mathcal{D}(A) \\ A u = \lim_{n \rightarrow \infty} A u_n \end{array} \right.$$

c) If  $(T_t)_{t \geq 0}$  is another Feller semigroup with generator  $(A, \mathcal{D}(A))$ , then  $P_t = T_t$ .

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The resolvent

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Let  $(P_t)_{t \geq 0}$  be a Feller semigroup. Then the integral

$$U_\alpha u(x) := \int_0^\infty e^{-\alpha t} P_t u(x) dt, \quad \alpha > 0, x \in \mathbb{R}^d, \\ u \in \mathcal{B}_b(\mathbb{R}^d)$$

exists and defines a linear operator. Indeed

$$|U_\alpha u(x)| \leq \int_0^\infty |P_t u(x)| e^{-\alpha t} dt$$

$$\leq \|u\|_\infty \int_0^\infty e^{-\alpha t} dt$$

$$= \|u\|_\infty \frac{1}{\alpha}.$$

Def. The operator  $U_\alpha$  so defined is called the  $\alpha$  potential operator associated with  $P_t$ .  $\square$

Recall on spectral theory.

Let  $T$  be a bounded linear operator on a Banach space  $X$ . The resolvent set of  $T$  is the set  $\rho(T)$  of (complex) numbers  $\lambda$  s.t.  $(\lambda I - T)$  is invertible, i.e., there exists an inverse operator (which is bounded).

The resolvent of  $T$  is defined as

$$R_\lambda(T) := (\lambda I - T)^{-1}$$

The spectrum of  $T$  is the set  $\sigma(T)$  of (complex) numbers not belonging to the resolvent set, i.e., s.t. the operator  $(\lambda I - T)$  is not invertible.

If  $T$  is unbounded it is the same definition

$$e) \quad \alpha U_\alpha u - u = \alpha \int_0^\infty e^{-\alpha t} T_t u \, dt - u = \alpha \int_0^\infty e^{-\alpha t} (T_t u - u) \, dt \\ = \int_0^\infty e^{-t} (T_t u - u) \, dt$$

Proposition. Let  $(P_t)_{t \geq 0}$  be a Feller semigroup with 2-potential operator  $(U_\alpha)_{\alpha > 0}$  and generator  $(A, \mathcal{D}(A))$ .

- $\alpha U_\alpha$  is conservative whenever  $(P_t)_{t \geq 0}$  is conservative.
- $\alpha U_\alpha$  is a contraction on  $\mathcal{B}_b$ .
- $\alpha U_\alpha$  is positivity preserving.
- $\alpha U_\alpha$  has the Feller property.
- $\alpha U_\alpha$  is strongly continuous on  $C_0$ , i.e.,  $\lim_{\alpha \rightarrow \infty} \|\alpha U_\alpha u - u\|_\infty = 0$ .
- $(U_\alpha)_{\alpha > 0}$  is the resolvent: for all  $\alpha > 0$ ,  $\alpha 1 - A$  is invertible with a bounded inverse  $U_\alpha = (\alpha 1 - A)^{-1}$ .
- Resolvent equation  $(U_\alpha u - U_\beta u) = (\beta - \alpha) U_\beta U_\alpha u$  for all  $\alpha, \beta > 0$  and  $u \in \mathcal{B}_b$ . In particular  $\lim_{\beta \rightarrow \alpha} \|U_\alpha u - U_\beta u\| = 0$  for all  $u \in \mathcal{B}_b$ .
- There is a 1-to-1 correspondence between  $(P_t)_{t \geq 0}$  and  $(U_\alpha)_{\alpha > 0}$ .
  - The resolvent  $(U_\alpha)_{\alpha > 0}$  is sub-Markovian iff  $P_t$  is sub-Markovian.

Proof

$$a) \alpha U_\alpha 1 = \alpha \int_0^\infty e^{-\alpha t} dt = 1$$

$$b) |\alpha U_\alpha u| \leq \alpha \int_0^\infty e^{-\alpha t} |P_t u| dt$$

$$\leq \|u\|_\infty \alpha \int_0^\infty e^{-\alpha t} dt$$

$$c) \text{ let } u \geq 0. \quad \alpha \int_0^\infty e^{-\alpha t} \underbrace{P_t u}_{\geq 0} dt \geq 0$$

since  $P_t$  is positivity preserving

$$d) \alpha U_\alpha u(x) = \alpha \int_0^\infty e^{-\alpha t} \underbrace{P_t u(x)}_{|P_t u(x)| \leq \|u\|_\infty} dt$$

$$|P_t u(x)| \leq \|u\|_\infty$$

Hence

$$\text{and } P_t u \in C_0(\mathbb{R}^d)$$

$$\alpha U_\alpha u(x) \in C_0(\mathbb{R}^d) \text{ by DCT conv.}$$

e) Note that

$$\begin{aligned} \|\alpha U_\alpha u - u\|_\infty &= \left\| \alpha \int_0^\infty e^{-\alpha t} P_t u(x) dt - u(x) \right\| \\ &= \left\| \int_0^\infty \alpha e^{-\alpha t} (P_t u(x) - u(x)) dt \right\| \\ &\leq \int_0^\infty \alpha e^{-\alpha t} \|P_t u - u\| dt \end{aligned}$$



$$= \int_0^\infty e^{-\alpha s} \|P_{\alpha/2} u - u\|_p ds \quad \|P_{\alpha/2} u - u\| \leq 2u$$

By GNV

$$\rightarrow 0 \quad \alpha \rightarrow +\infty.$$

ℓ) We prove first that  $U_\alpha u \in \mathcal{D}(A)$  for every  $u \in C_\infty(\mathbb{R}^d)$  and that  $A U_\alpha u = \alpha U_\alpha u - u$  (hence  $(\alpha I - A) U_\alpha u = u$ ) for all  $u \in C_\infty(\mathbb{R}^d)$

$$A U_\alpha u = \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (P_\varepsilon U_\alpha u - U_\alpha u) =$$

$$= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \int_0^\infty e^{-\alpha t} (P_{t+\varepsilon} u - P_t u) dt$$

$t+\varepsilon = r$

$$= \lim_{\varepsilon \rightarrow 0} \left[ \frac{1}{\varepsilon} \int_\varepsilon^\infty e^{-\alpha(r-\varepsilon)} P_r u dr - \frac{1}{\varepsilon} \int_0^\infty e^{-\alpha r} P_r u dr \right]$$

$$= \lim_{\varepsilon \rightarrow 0} \left[ \frac{e^{\alpha\varepsilon}}{\varepsilon} \int_0^\infty e^{-\alpha r} P_r u dr - \frac{e^{\alpha\varepsilon}}{\varepsilon} \int_0^\varepsilon e^{-\alpha(r-\varepsilon)} P_r u dr - \frac{1}{\varepsilon} \int_0^\infty e^{-\alpha r} P_r u dr \right]$$

$$= \lim_{\varepsilon \rightarrow 0} \left[ \underbrace{\frac{e^{\alpha\varepsilon}}{\varepsilon}}_{\rightarrow \alpha} \int_0^\infty e^{-\alpha r} P_r u dr - \underbrace{\frac{e^{\alpha\varepsilon}}{\varepsilon} \int_0^\varepsilon e^{-\alpha(r-\varepsilon)} P_r u dr}_{\rightarrow u} \right]$$

$$= \alpha \int_0^\infty e^{-\alpha r} P_r u dr - u$$

$$= \alpha U_\alpha u - u$$

proved in previous theorem

for  $u \in \mathcal{D}(A)$   
 Now we prove that  $U_\alpha A u = A U_\alpha u$  and thus

$$u = (\alpha I - A) U_\alpha u = \underbrace{\alpha U_\alpha u} - \underbrace{A U_\alpha u} \quad u \in \mathcal{D}(A)$$

$$= U_\alpha \alpha u - U_\alpha A u$$

$$= U_\alpha (\alpha I - A) u$$

i.e.  $U_\alpha$  is the left and right inverse of  $\alpha I - A$ .

Since  $U_\alpha$  is bounded  $\alpha I - A$  has a bounded inverse,

i.e.,  $\alpha > 0$  is in the resolvent set and  $U_\alpha$  is the resolvent.

$$U_\alpha A u = \int_0^\infty e^{-\alpha t} P_t A u dt = \lim_{R \rightarrow \infty} \int_0^R A P_t u e^{-\alpha t} dt = \lim_{R \rightarrow \infty} A \int_0^R P_t u e^{-\alpha t} dt$$

Since  $A$  is closed  
 $\lim_{R \rightarrow \infty} \int_0^R P_t u e^{-\alpha t} dt \in \mathcal{D}(A)$   
 and  
 $\lim_{R \rightarrow \infty} A = A \lim_{R \rightarrow \infty}$

g) Let  $\alpha, \beta > 0$  and  $u \in \mathcal{D}(A)$ . Then

$$U_\alpha U_\beta u = U_\beta U_\alpha u$$

↳ Fubini

thus

$$\begin{aligned} U_\alpha u - U_\beta u &= (\beta I - A) U_\beta U_\alpha u - (\alpha I - A) U_\alpha U_\beta u \\ &= (\beta - \alpha) U_\beta U_\alpha u. \end{aligned}$$

hence, since  $\alpha U_\alpha$  is contractive,

$$\|U_\alpha u - U_\beta u\|_\infty = \frac{|\beta - \alpha|}{\beta \alpha} \|\alpha U_\alpha \beta U_\beta u\|_\infty$$

$$\leq \left| \frac{1}{\alpha} - \frac{1}{\beta} \right| \|u\|_\infty \xrightarrow{\beta \rightarrow \alpha} 0.$$

b) This follows by uniqueness of Laplace transform, i.e., if  $f$  and  $g$  are Laplace transformable for some  $\lambda > \lambda_0$  and  $\tilde{f}(\lambda) = \tilde{g}(\lambda)$  for all  $\lambda > \lambda_0$ . Then  $f(t) = g(t)$  a.e.

It follows that, if  $f$  and  $g$  are continuous, then  $f = g$  for all  $t \geq 0$ .

Hence  $P_t u(x)$  is uniquely determined by  $U_\alpha u(x)$ .

i) Take  $u \in \mathcal{B}_b(\mathbb{Q}^d)$  s.t.  $0 \leq u \leq 1$ . Then

$$0 \leq P_t u \leq 1 \rightarrow \underbrace{\alpha \int_0^\infty e^{-\alpha t} P_t u \, dt}_{0.5 \downarrow 0.5}$$

$0.5$

$\leq 1$

The converse can be proved by resorting to completely monotone functions — see References.

Example: Bochner-Martin.

For  $d=1$   
 $u \in \mathcal{B}_b$

$$\begin{aligned} U_\alpha u(x) &= \int_0^\infty e^{-\alpha t} P_t u(x) dt \\ &= \int_{\mathbb{R}} \underbrace{\frac{e^{-\sqrt{2\alpha}|y|}}{\sqrt{2\alpha}}}_{\text{resolvent density}} u(x-y) dy \end{aligned}$$

For  $d \geq 2$   
 $u \in \mathcal{B}_b(\mathbb{R}^d)$

Resolvent density:

$$\frac{1}{\Gamma(\frac{d}{2})} \left( \frac{\sqrt{2\alpha}}{2|y|} \right)^{\frac{d}{2}-1} K_{\frac{d}{2}-1}(\sqrt{2\alpha}|y|) \quad d \geq 1$$

↓  
Bessel function.

Let us use the resolvent's properties to determine the domain of the generator of BM.

For  $d=1$ :  $\mathcal{D}(A) = C_c^\infty(\mathbb{R}) = \{u \in C_c : u, u', u'' \in C_c(\mathbb{R})\}$

Since  $U_\alpha$  is the inverse of  $\alpha I - A$  we have that

$U_\alpha(C_c(\mathbb{R})) = \mathcal{D}(A)$ . Hence, take  $u \in \mathcal{D}(A)$ , we have that  $u = U_\alpha f$  for some  $f \in C_c(\mathbb{R})$ .

Using the representation of the resolvent:

$$U_\alpha f(x) = \frac{1}{\sqrt{2\alpha}} \int_{-\infty}^{\infty} e^{-\sqrt{2\alpha}|y-x|} f(y) dy$$

$f(x) = g \in \mathcal{D}$   
 with  $\alpha u'' = f$   
 $\mathcal{D}(f) \ni x \mapsto u \mapsto \alpha u'' = f$   
 $\mathcal{D}(f) \ni x \mapsto u \mapsto \alpha u'' = f$   
 $f(x) = g \in \mathcal{D}$   
 with  $\alpha u'' = f$   
 $\mathcal{D}(f) \ni x \mapsto u \mapsto \alpha u'' = f$

Taking derivative w.r.t  $x$ :

$$\begin{aligned}\partial_x V_x f(x) &= \frac{\sqrt{x}}{\sqrt{x}} \int_{-\infty}^{+\infty} e^{-\sqrt{x}(y-x)} \sqrt{x}(y-x) f(y) dy \\ &= \int_x^{+\infty} e^{-\sqrt{x}(y-x)} f(y) dy \\ &\quad - \int_{-\infty}^x e^{-\sqrt{x}(x-y)} f(y) dy\end{aligned}$$

$$\partial_x^2 V_x f(x) = \sqrt{x} \int_{-\infty}^{+\infty} e^{-\sqrt{x}(y-x)} f(y) dy - 2 f(x)$$

(\*) The method: let  $\phi \in (a,b) \subset \mathbb{R}$  be an open interval and  $u: (a,b) \times X \rightarrow \mathbb{R}$  a function s.t.

-  $y \mapsto u(x,y)$  is in  $L^1_\mu$  for every fixed  $x \in (a,b)$

-  $x \mapsto u(x,y)$  is differentiable  $\mu$ -a.e.  $y \in \mathbb{R}$

-  $|\partial_x u(x,y)| \leq w(y)$  for all  $(x,y) \in (a,b) \times \mathbb{R}$  and  $w \in L^1_\mu$   
 $\mu$ -a.e.  $y \in \mathbb{R}$

Let the function

$$(a,b) \ni x \mapsto \int u(x,y) \mu(dy)$$

is differentiable and

$$\partial_x \int u(x,y) \mu(dy) = \int \partial_x u(x,y) \mu(dy)$$

We see here that  
 $y \mapsto \frac{e^{-|y-n|\sqrt{2\alpha}}}{\sqrt{2\alpha}} f(y)$  is in  $L^1$  for every  
 $n \in (a, b)$ ,  $a, b$  arbitrary.

(9)  $n \mapsto \frac{e^{-|y-n|\sqrt{2\alpha}}}{\sqrt{2\alpha}} f(y)$  is diff. for a.a.  $y$   
 $y \in \mathbb{R}$  (by Lebesgue's theorem on diff. under the integral).

~~$$\frac{\partial}{\partial n} \left( \frac{e^{-|y-n|\sqrt{2\alpha}}}{\sqrt{2\alpha}} f(y) \right) = -\sqrt{2\alpha} \frac{e^{-|y-n|\sqrt{2\alpha}}}{\sqrt{2\alpha}} f(y)$$~~

Note now that

$$\partial_n^2 U_\alpha f(n) = 2\alpha U_\alpha f(n) - 2 f(n)$$

and since  $U_\alpha f(n) \in C_0(\mathbb{R})$  ( $f(n) \in C_0(\mathbb{R})$ ) then  
 $\partial_n^2 U_\alpha f(n) \in C_0(\mathbb{R})$ . It follows that

$$\mathcal{D}(A) \subset C_0^2(\mathbb{R})$$

Since we proved that, for every  $u \in \mathcal{D}(A)$  we have  
 $\partial_n^2 u \in C_0(\mathbb{R})$

$$\partial_n u = \int_{\mathbb{R}} \frac{e^{-|y-n|\sqrt{2\alpha}}}{\sqrt{2\alpha}} f(y) dy = \int_{\mathbb{R}} \frac{e^{-|y-n|\sqrt{2\alpha}}}{\sqrt{2\alpha}} f(y) dy$$

$\xrightarrow{\text{C}_0 \text{ by Lebesgue}}$

# — Hille - Yosida —

Theorem. Let  $(Z, \|\cdot\|)$  be a Banach space. A linear operator  $(A, \mathcal{D}(A))$  is the infinitesimal generator of a strongly continuous contraction semigroup if, and only if,

a)  $\mathcal{D}(A)$  is dense in  $Z$

b)  $A$  is dissipative i.e.  $\forall \lambda > 0, u \in \mathcal{D}(A)$

$$\rightarrow \| \lambda u - Au \| \geq \lambda \| u \|$$

c)  $R(\lambda I - A) = Z$  for some (or all)  $\lambda > 0$

↓  
range

b) + c) means **maximally dissipative**

b) implies  $\lambda - A$  injective c) implies surjectivity

Thus it hard to be used, but:

$$b) + c) \text{ implies } (0, \infty) \subset \rho(A)$$

Lemma. Let  $(P_t)_{t \geq 0}$  be a Feller semigroup with generator  $(A, \mathcal{D}(A))$ . The  $A$  satisfies the positive maximum principle

$$u \in \mathcal{D}(A), u|_{\partial \Omega} = 0, u|_{\Omega} \geq 0 \rightarrow Au|_{\Omega} \leq 0$$

↓  
If  $A = \Delta_{\Omega}$  this looks familiar.

Def.



$$u = u^+ - u^-$$

$$u^+ - u = \underbrace{u^-}_{\geq 0}$$

let  $u \in D(A)$  and assume that  $u(x_0) = \inf_{x \in \Omega} u(x) \geq 0$ .

Then,  $\forall x \in \Omega$ ,  $u - u \geq 0$  and  $\forall t > 0$ ,  $P_t(u^+ - u) \geq 0$  by positivity preserving property. This means that

$$P_t u^+ \geq P_t u, \quad \forall x$$

Hence

$$\frac{1}{t} (P_t u(x_0) - u(x_0)) \leq \frac{1}{t} (P_t u^+(x_0) - \underbrace{u(x_0)}_{= u^+(x_0)})$$

since  $u(x_0)$  is positive.

$$= \frac{1}{t} (P_t u^+(x_0) - u^+(x_0))$$

$$\leq \frac{1}{t} (||u^+|| - u^+(x_0)) = 0.$$

$$\hookrightarrow \text{since } u^+(x_0) = \sup_{x \in \Omega} u(x)$$

\* 6) of H-Y lemma

Further

Lemma. Assume that  $(A, D(A))$  satisfies the condition  $D(A)$  is dense in  $\mathcal{H}$  and  $R(\lambda I - A) = \mathcal{H}$ . Suppose also that the positive maximum principle is satisfied. Then  $(A$  is dissipative)  $\Rightarrow$  and, in particular,  $(\lambda I - A)$  is invertible and both the resolvent  $U_\lambda$  and the semigroup generated are positivity preserving.

(Here  $\mathcal{H} = C(\Omega)$ )



Let  $u \in \mathcal{Q}(A)$  and  $x_0$  s.t.  $|u(x_0)| = \inf |u(x)|$   
 Assume  $u(x_0) \geq 0$  (otherwise consider  $-u$ )  
 Then, for all  $\lambda > 0$

$$\begin{aligned} \| \lambda u - A u \|_{\infty} &\geq \lambda u(x_0) - \underbrace{A u(x_0)}_{\geq 0 \text{ by PMP}} \\ &\geq \lambda u(x_0) \\ &= \lambda \|u\|_{\infty} \end{aligned}$$

This is b) of Hille-Yosida. b) & c) imply  
 invertibility of  $(\lambda - A)$ . We prove for preserving.

Recall that  $T_t$  is Markov iff  $U_t$  is Markov  
 Hence suppose  $v \geq 0$  and consider  $U_t v$ .

If  $U_t v$  does not attain the inf, then  
 $U_t v \geq 0$  or  $U_t v$  is continuous or vanishes at infinity.

If  $U_t v$  attains the inf, then  $U_t v(x_0) \leq 0$   
 since  $U_t v$  vanishes at infinity.

Hence  $U_t v$  is a function in  $\mathcal{Q}(A)$  and non-positive  
~~minimum~~ minimum in  $x_0$ . If we apply the  
 Feller PMP (valid for a positive function)  
 to the function  $h = -u$ , we obtain

$$h \in \mathcal{Q}(A) \quad h(x_0) = \inf_x h(x) \leq 0 \rightarrow A h(x_0) \geq 0$$



which is called minimum principle. Apply this  
to  $U_\lambda v$  to get

$$\lambda U_\lambda v(x_0) - v(x_0) = A U_\lambda v(x_0) \geq 0 \quad \square$$

Hence  $\lambda U_\lambda v(x_0) \geq \inf_x \lambda U_\lambda v(x)$

$$= \lambda U_\lambda v(x_0)$$

$\square$

$$\geq v(x_0) \geq 0$$

Hence  $U_\lambda v \geq 0$  since  $x_0$  is a minimum

The last result we will need on generators is the following.

Thm. Let  $(P_t)_{t \geq 0}$  be a Feller semigroup generated by  $(A, \mathcal{D}(A))$ . Then

$$\mathcal{D}(A) = \left\{ u \in C_\infty(\mathbb{R}^d) : \exists p \in C_\infty(\mathbb{R}^d) : \lim_{t \rightarrow 0} \frac{P_t u(x) - u(x)}{t} = p(x), \forall x \in \mathbb{R}^d \right\}$$


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Remark on surjectivity.

c) is equiv. to  $(\lambda - A)\mathcal{D} \subset C_\infty$

is dense. Indeed, let  $u \in C_\infty$  then  $\exists u_n \subset (\lambda - A)\mathcal{D}$  s.t.

$$u_n = (\lambda - A)h_n \text{ and}$$

$$(\lambda - A)h_n \rightarrow u \quad \nearrow \quad \parallel$$

Since  $A$  closed, then  $(\lambda - A)h_n \rightarrow (\lambda - A)h$  where  $h = \lim_n h_n$